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SECJ3623: Mobile Application Programming

Final Project Report

<Study Companion + >

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1. INTRODUCTION

Digital technology has become essential in modern education. However, many students still depend on notebooks, paper homework, and messaging apps to manage their studies. This approach often causes:

- forgotten assignments
- unclear learning schedules
- limited teacher–parent communication
- poor tracking of student progress

StudyCompanion+ is developed as a mobile solution to centralize learning activities into one platform. The application supports daily academic management such as tracking homework, reminders, announcements and monitoring student's performance.

By digitizing these processes, the system provides a more structured, organized, and convenient learning experience for students, teachers, and parents.

1.1 Problem Statement

In many educational environments, the management of learning activities is still fragmented and inefficient. Homework assignments are often communicated verbally or written on whiteboards, increasing the likelihood that students may forget or misunderstand the tasks. Learning materials may be distributed through multiple channels such as emails or messaging applications, which can create confusion and inconsistency.

Furthermore, teachers lack a convenient method to monitor individual student performance continuously. Evaluating progress typically requires manual checking of exercise books or paper-based records, which is time-consuming and prone to errors. Parents also face difficulties obtaining real-time information about their child's academic development, limiting their ability to intervene early when problems arise.

These issues highlight the need for a centralized and structured system that can integrate learning resources, task management, and progress monitoring into one digital platform. Without such a system, both teaching and learning processes remain inefficient and uncoordinated.

1.2 Project Objectives

The primary objective of StudyCompanion+ is to design and develop a mobile application that improves classroom management and enhances learning experience for all stakeholders. The project seeks to provide a centralized platform that simplifies the distribution of materials, assignment of tasks, and monitoring of student progress.

Specifically, the system aims to enable students to access study materials and complete assignments conveniently through their mobile devices. It also allows teachers to manage coursework digitally and evaluate performance more efficiently. Additionally, the system provides parents with visibility into their child's academic achievements, encouraging better collaboration between home and school. Through these objectives, StudyCompanion+ strives to promote a more organized, transparent, and effective learning environment.

1.3 Project Perspective

StudyCompanion+ is developed as a mobile client–server system that supports multiple user roles. The architecture is designed to ensure smooth communication between the mobile application, backend services, and database storage. Each component plays a specific role in maintaining system reliability and efficiency.

The mobile application serves as the main interface through which users interact with the system. All user requests are processed by the backend server, which performs the necessary logic and communicates with the database to store or retrieve information. This layered structure ensures scalability, maintainability, and consistent performance.

2. STAKEHOLDER AND USERS

2.1 Stakeholders

Stakeholders include individuals or organizations that influence or are affected by the system, while users refer to those who interact directly with the mobile application. Understanding their roles helps ensure that system features align with practical requirements and improve overall usability.

Stakeholder	Interest in the System	Responsibilities	Expected Benefits
School Management	Ensure smooth academic operations and performance monitoring	Approve system adoption, oversee usage policies	Better visibility of school performance and digital management
Teachers	Deliver lessons, assign tasks, evaluate students	Upload materials, assign homework, review results	Easier task management and reduced paperwork
Parents/Guardians	Monitor child's academic progress	Track performance, respond to notifications	Improved communication with teachers and better awareness of child's learning
Students	Access learning resources and complete assignments	Participate in learning activities	Organized study schedule and improved performance
IT/Developer Team	Maintain system reliability and updates	Develop, test, and support the application	Stable and scalable application performance
Faculty/School Board	Evaluate system effectiveness and outcomes	Provide strategic direction and approval	Data-driven academic improvement

2.2 Users

Users are individuals who directly interact with StudyCompanion+ through the mobile application. Each user role is designed with specific permissions and functions to ensure proper system operation.

User Type	Access Level	Main Activities	System Features Used
Student	Basic access	Study, complete tasks	View tasks, submit homework, view progress/performance
Teacher	Medium–High access	Manage teaching materials and assessments	Manage classrooms, create tasks, grade students, monitor performance
Parent	Limited access	Monitor child's progress	View reports, receive notifications
Admin	Full access	Manage users and system settings	Create accounts, manage database, monitor usage

2.3 User Needs

By aligning system features with actual user requirements, the application enhances usability, productivity, and satisfaction for all stakeholders. It ensures that StudyCompanion+ delivers user-centered functionality as desired.

User	Responsibilities	Needs / Problems Faced	How StudyCompanion+ Supports
Student	Complete assignments and learning tasks	Forget homework, lose assignments, lack organization	Centralized dashboard, reminders, easy access to materials
Teacher	Deliver lessons and assess students	Manual grading, paperwork, tracking difficulties	Digital assignment management and progress monitoring
Parent	Support child's education	Lack of performance visibility	Real-time progress reports and notifications
Admin	Maintain system operations	User management complexity	Centralized control panel

3 UML USE CASE DIAGRAM

The Use Case Diagram illustrates the functional interactions between external users and the StudyCompanion+ system. It provides a high-level overview of how different actors communicate with the mobile application to perform their respective tasks. The diagram identifies four main actors, namely Student, Teacher, Parent, and Admin, each representing a specific role within the learning environment.

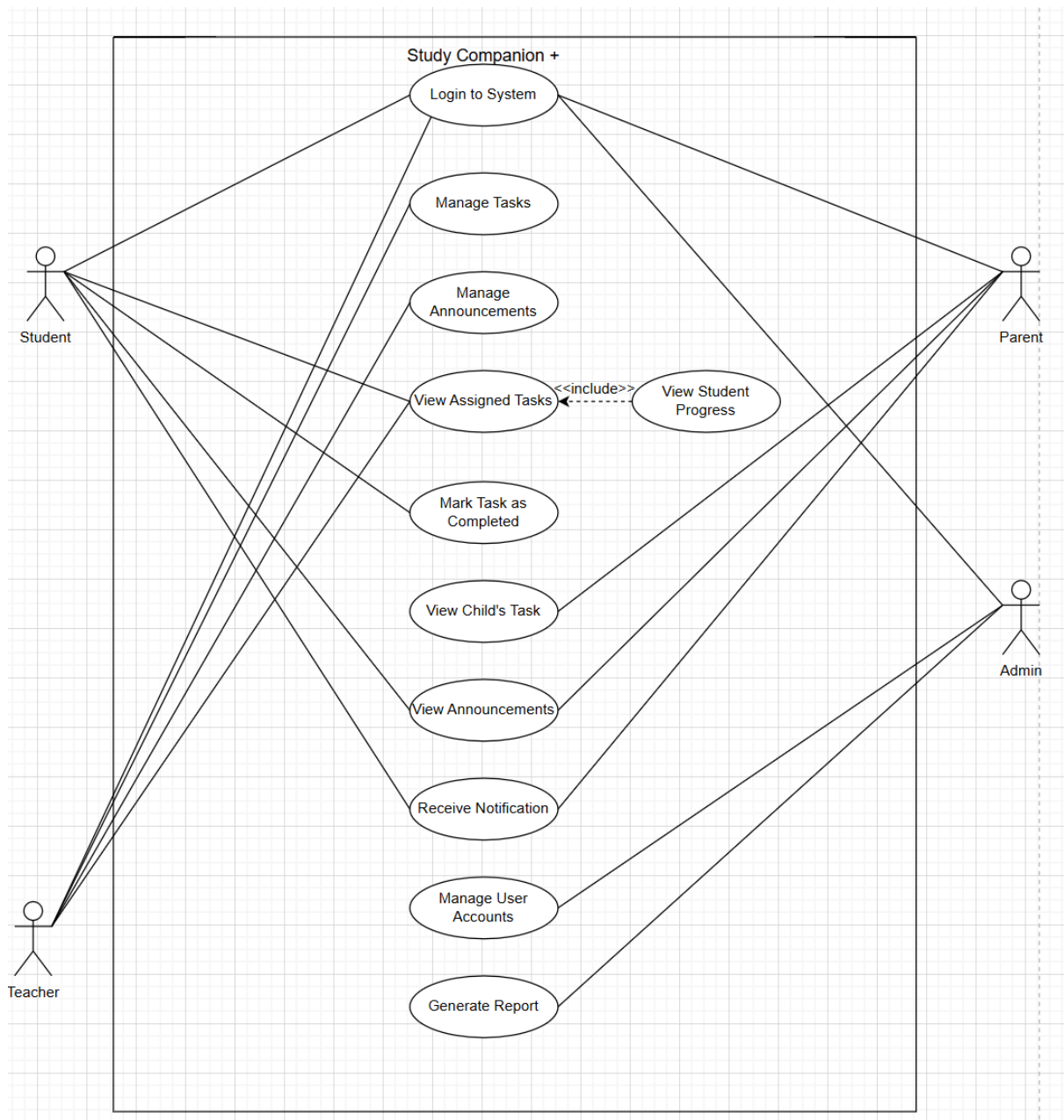


Figure 3.0 USE CASE DIAGRAM for Study Companion +

3.1 Actor Definition

StudyCompanion+ consists of four primary actors: Student, Teacher, Parent, and Admin.

Actor	Type	Description
Student	Primary Actor	A learner who uses the mobile application to manage daily academic activities and complete assigned tasks.
Teacher	Primary Actor	An educator responsible for managing coursework, assignments, and monitoring student learning outcomes.
Parent	Secondary Actor	A guardian who monitors their child's academic progress and receives updates from the school.
Admin	Supporting Actor	A system administrator responsible for maintaining system operations and managing user accounts.

3.2 Use Case Summary Table

The use case summary table provides an overview of all functional requirements supported by StudyCompanion+. Each use case corresponds to a specific interaction between system actors and the application. The use cases ensure that students can manage learning tasks, teachers can administer classroom activities, parents can monitor progress, and administrators can maintain system operations effectively.

Use Case ID	Use Case Name	Primary Actor(s)	Brief Description
UC01	Login to System	Student, Teacher, Parent, Admin	Allows users to authenticate and securely access the system using valid credentials.
UC02	Manage Tasks	Teacher	Enables teachers to create, edit, assign, or delete homework and learning tasks for students.
UC03	Manage Announcements	Teacher	Allows teachers to post and update announcements or important notices for users.
UC04	View Assigned Tasks	Student, Teacher	Displays the list of tasks or assignments assigned to students for completion.

Use Case ID	Use Case Name	Primary Actor(s)	Brief Description
UC05	Mark Task as Completed	Student	Allows students to update the task status after completing assigned work.
UC06	View Student Progress	Teacher, Parent	Shows academic performance and task completion records for monitoring purposes.
UC07	View Child's Task	Parent	Enables parents to view tasks specifically assigned to their child.
UC08	View Announcements	Student, Teacher, Parent	Allows users to read posted announcements and updates from teachers or school management.
UC09	Receive Notification	Student, Teacher, Parent, Admin	Sends reminders or alerts regarding tasks, announcements, or system updates.
UC10	Manage User Accounts	Admin	Allows the administrator to create, modify, or delete user accounts and manage access rights.
UC11	Generate Report	Admin	Generates summary reports related to system usage, student performance, or academic records.

4 SYSTEM FLOW

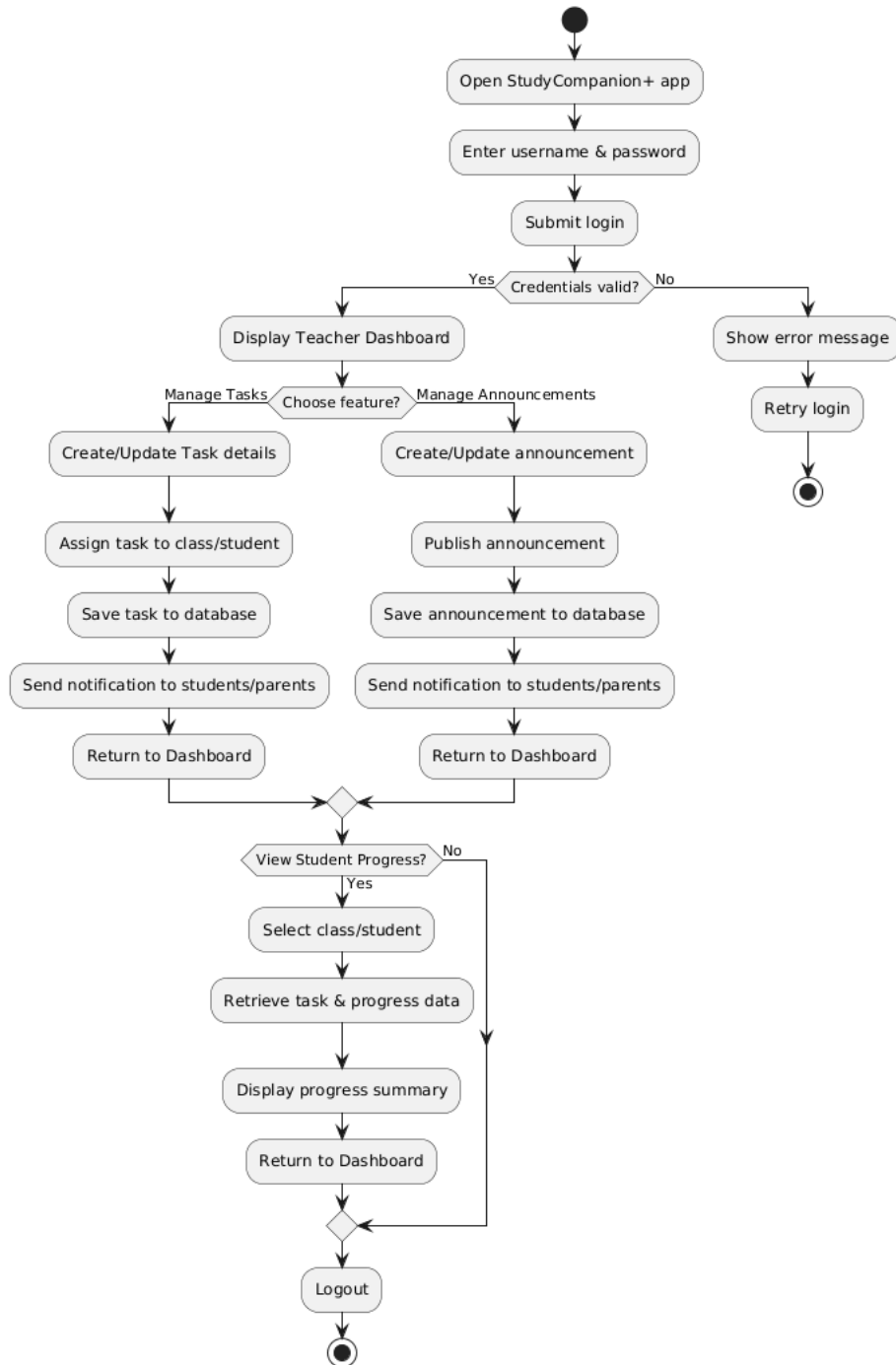
The system flow of StudyCompanion+ illustrates the sequence of interactions between users and the mobile application based on role-specific activities. Each flow begins with user authentication, where credentials are validated before access to the respective dashboard is granted. Upon successful login, the system routes the user to functions permitted by their assigned role.

Teachers perform task and announcement management, where created or updated data are stored in the database and notifications are dispatched to relevant users. Students retrieve assigned tasks, update completion status, and access announcements, with all actions recorded to maintain accurate progress tracking. Parents access monitoring functions to review their child's tasks, progress, and announcements without modifying academic records.

All user actions are processed by the application layer and synchronized with the database to ensure data consistency and real-time updates. The activity diagrams provide a visual representation of these operational workflows and depict the control flow between user actions and system responses.

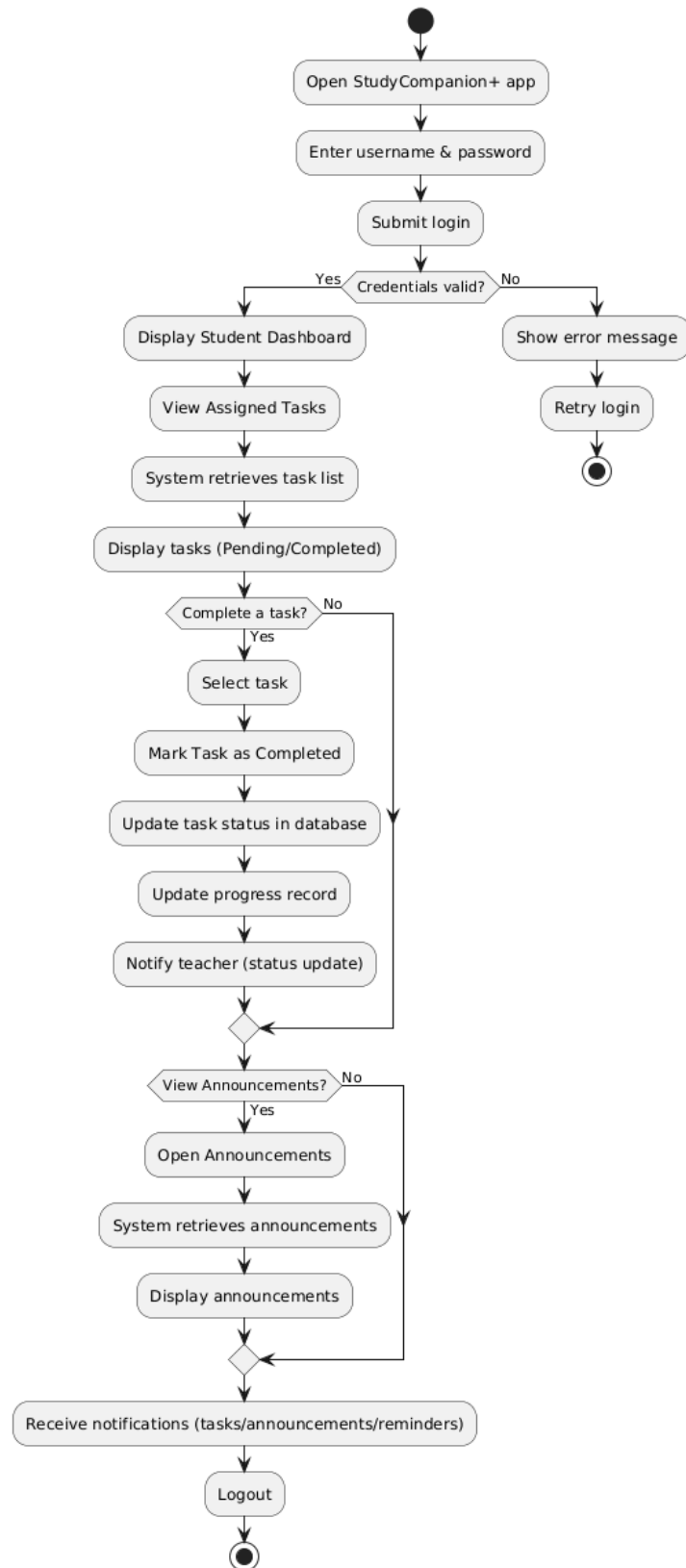
4.1 Activity Diagram (Teacher)

Activity Diagram - Teacher (StudyCompanion+)



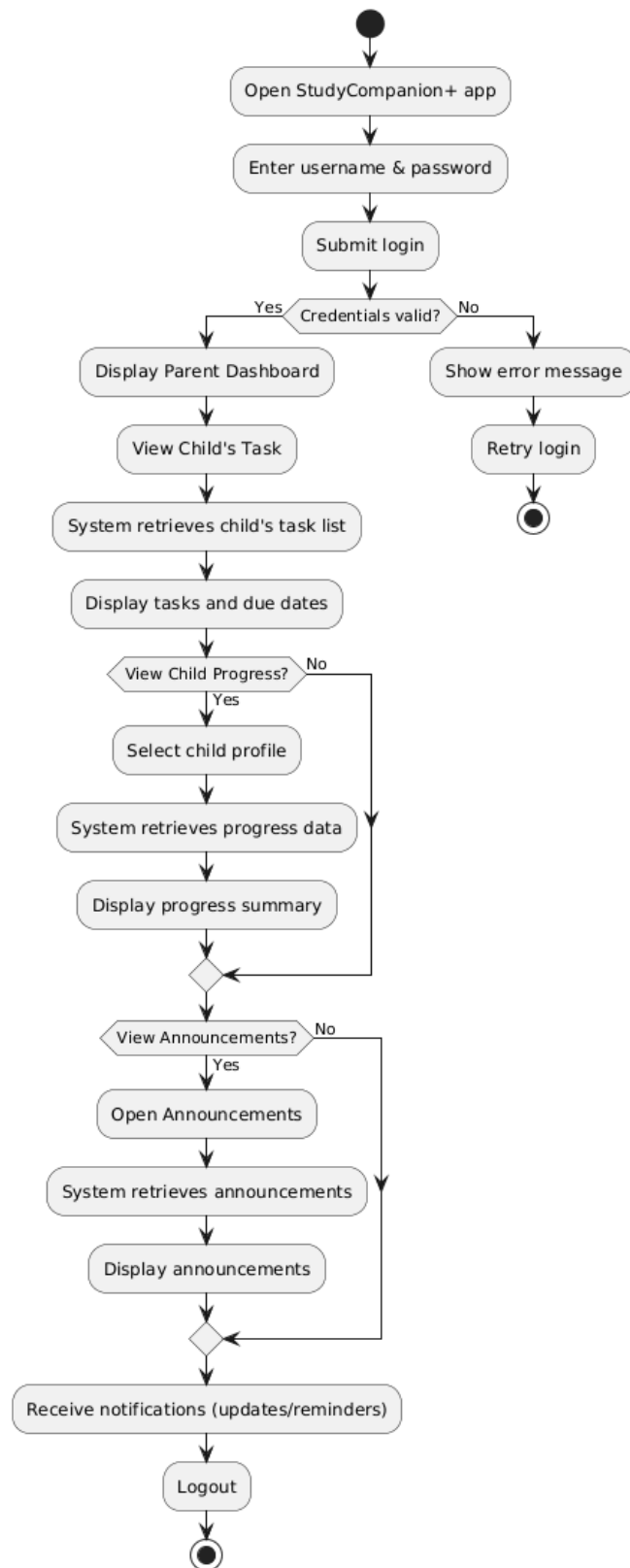
4.2 Activity Diagram (Student)

Activity Diagram - Student (StudyCompanion+)



4.3 Activity Diagram (Parent)

Activity Diagram - Parent (StudyCompanion+)



5 LOW FIDELITY UI DESIGN

Low-fidelity wireframes were created to define the basic layout, navigation structure, and component placement of the StudyCompanion+ mobile application. These wireframes focus on functionality and usability rather than visual styling. They help validate user flow before developing the high-fidelity interface

Screen Name	Low Fidelity UI Design
Login Screen (All Users)	<pre> +-----+ StudyCompanion+ ----- Select Role: [Student] [Parent] [Teacher] [Admin] Email / ID: [] Password: [] [Sign In] Forgot Password Sign Up +-----+ </pre>

Description: Provides role selection and authentication. Users choose their role and enter credentials to access their dashboard

Screen Name	Low Fidelity UI Design
Student Dashboard	<pre> +-----+ Header: Welcome, Student ----- Progress Card (Overall %) ----- Upcoming Reminders [Reminder 1] [Reminder 2] ----- Assigned Tasks List - Task A - Task B ----- Bottom Navigation Home Tasks Grades Profile +-----+ </pre>

Description: Displays overview of progress, reminders, and assigned tasks. Allows quick navigation to tasks and grades.

Screen Name	Low Fidelity UI Design
Task Management (Student)	<pre> +-----+ Header: My Tasks ----- Tabs: Assigned Pending Done ----- [Task Card] Title Due Date [Submit] ----- [Task Card] +-----+ Bottom Navigation +-----+ </pre>

Description: Shows task list with submission action. Students manage completion here.

Screen Name	Low Fidelity UI Design
Progress Screen	<pre> +-----+ Header: Progress Overview ----- Overall Progress % ----- Coursework Stats ----- Subject Performance List - Math - History ----- Recent Activity +-----+ </pre>

Description: Displays academic performance metrics and subject completion status.

Screen Name	Low Fidelity UI Design
Announcements/ Updates	<pre> +-----+ Header: Class Updates ----- Tabs: All Homework Exams ----- [Announcement Card] Title Short Description Date/Time ----- [Announcement Card] +-----+ Bottom Navigation +-----+ </pre>

Description: Lists school/class updates and notices. Users can view latest announcements.

Screen Name	Low Fidelity UI Design
Teacher Dashboard	<pre> +-----+ Header: Welcome, Teacher ----- Quick Actions [New Task] [Announce] [Log Grade] ----- Active Tasks - Task list ----- Class Overview Performance summary ----- Announcements +-----+</pre>

Description: Provides tools for managing tasks, announcements, and class monitoring

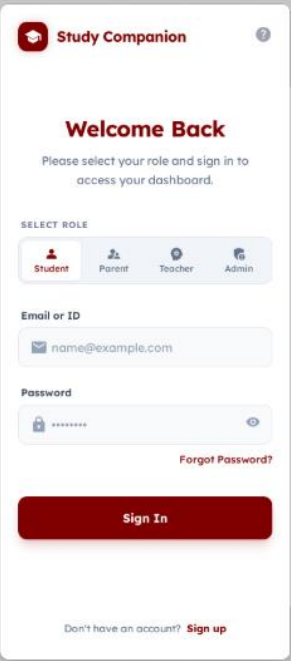
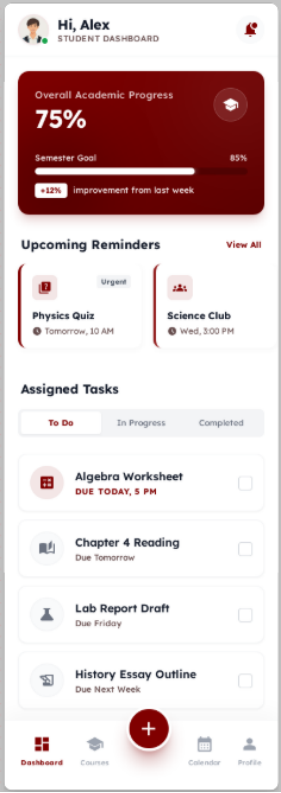
Screen Name	Low Fidelity UI Design
Parent Dashboard	<pre> +-----+ Header: Welcome, Parent ----- Child Profile Summary ----- Today's Tasks ----- Performance Overview ----- Announcements +-----+</pre>

Description: Allows parents to monitor child's tasks, performance, and school updates

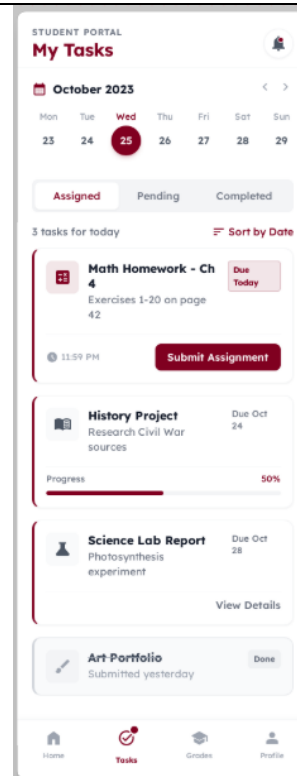
Screen Name	Low Fidelity UI Design
Admin Dashboard	<pre> +-----+ Header: Admin Console ----- Statistics Cards Students Teachers Reports ----- Quick Actions Manage Users Reports ----- Recent Activity Log +-----+ </pre>

Description: Provides system-level monitoring, user management, and reporting functions

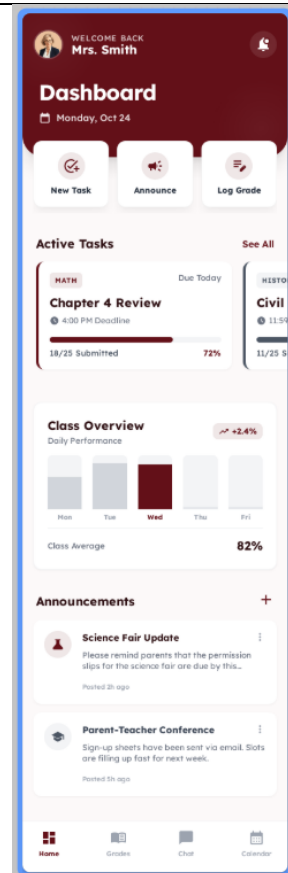
6 HIGH FIDELITY UI DESIGN

Screen Name	High Fidelity UI Design
Login	
Student Dashboard	

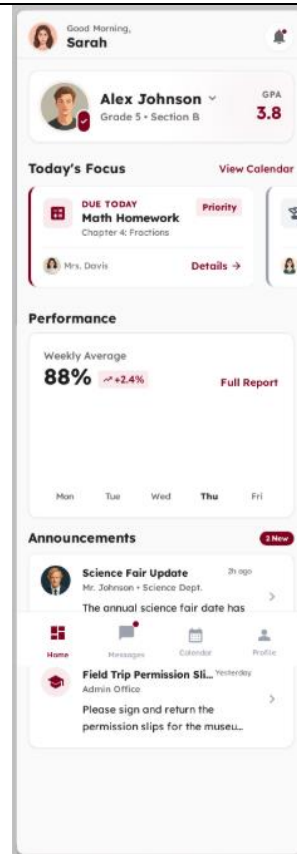
Student Task Management



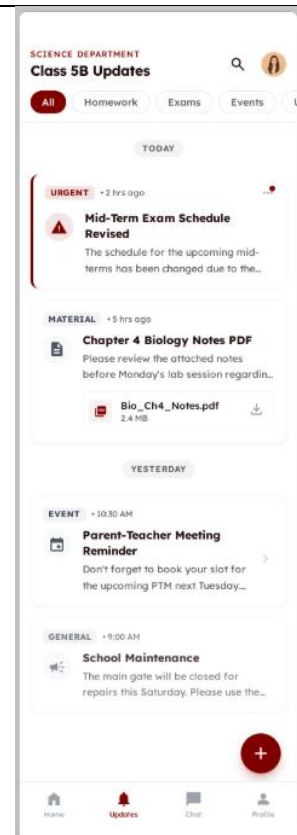
Teacher's Dashboard



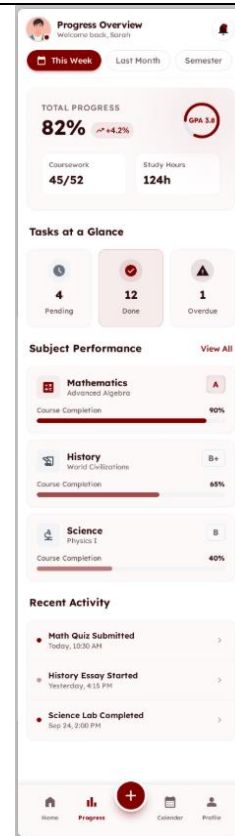
Parent's Dashboard



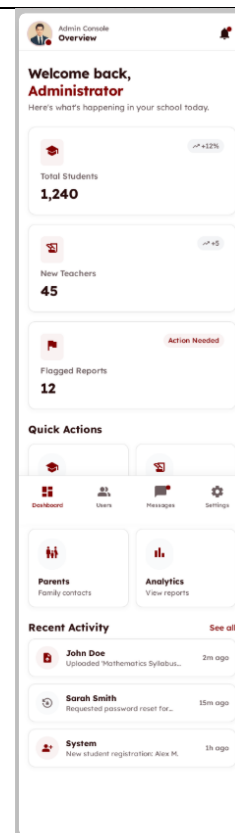
Announcement Screen



Progress View



Admin Dashboard



7 CONCLUSION

In conclusion, the StudyCompanion+ mobile application has been systematically designed to provide an integrated and structured platform for managing academic activities within the school environment. The proposed system addresses common challenges associated with manual task tracking, fragmented communication, and limited visibility of student progress by centralizing learning resources, assignments, announcements, and performance monitoring into a single mobile solution.

Through the identification of stakeholders and actors, the system requirements were clearly defined to ensure that each user group receives appropriate functionalities aligned with their responsibilities. The UML artefacts, including use case diagrams and activity diagrams, illustrate the interactions between users and the system, providing a clear representation of operational workflows and control flows. These models help validate functional behavior and ensure completeness of system processes.

Additionally, the low-fidelity wireframes established the basic layout and navigation structure, while the high-fidelity interface designs refined the visual presentation and user experience. This layered design approach improves usability and ensures that the system is intuitive, consistent, and accessible to all users.

Overall, the proposed StudyCompanion+ system offers a practical, efficient, and user-centered solution that enhances classroom management, improves communication among stakeholders, and supports better learning outcomes. The combination of structured analysis, UML modeling, and interface design provides a strong foundation for successful system implementation and deployment.